# Lecture Outline

- Linear models
- Estimate of the regression coefficients
    - Brute Force
    - Exact method
- Confidence intervals for the predictors estimates
- **Bootstrap**
- Evaluating significance of predictors
- How well we know the model $\hat{f}$

# Bootstrap

In the lack of active imagination, parallel universes and the likes, we need an alternative way of producing fake data set that resemble the parallel universes.

Bootstrapping is the practice of sampling from the observed data $(X,Y)$ in estimating statistical properties.

# Bootstrap

We first pick randomly a ball and replicate it. This is called sampling with replacement. We move the replicated ball to another bucket.

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We then randomly  pick another ball and again we replicate it. As before, we move the replicated ball to the other bucket.

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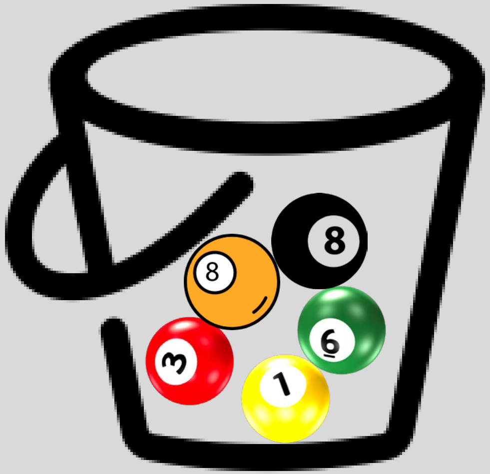
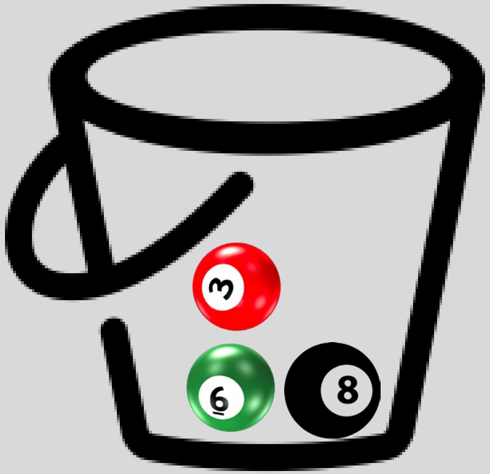
# Bootstrap

We repeat this process...

# Bootstrap

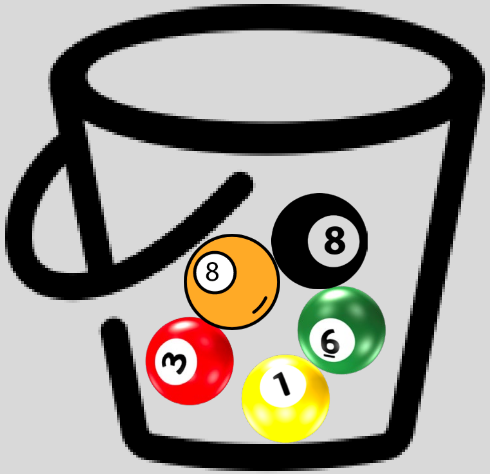
again

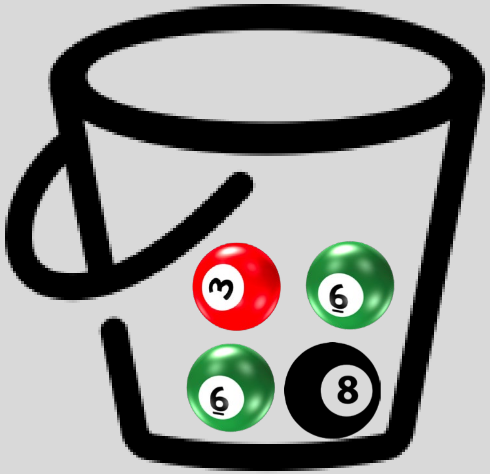


# Bootstrap

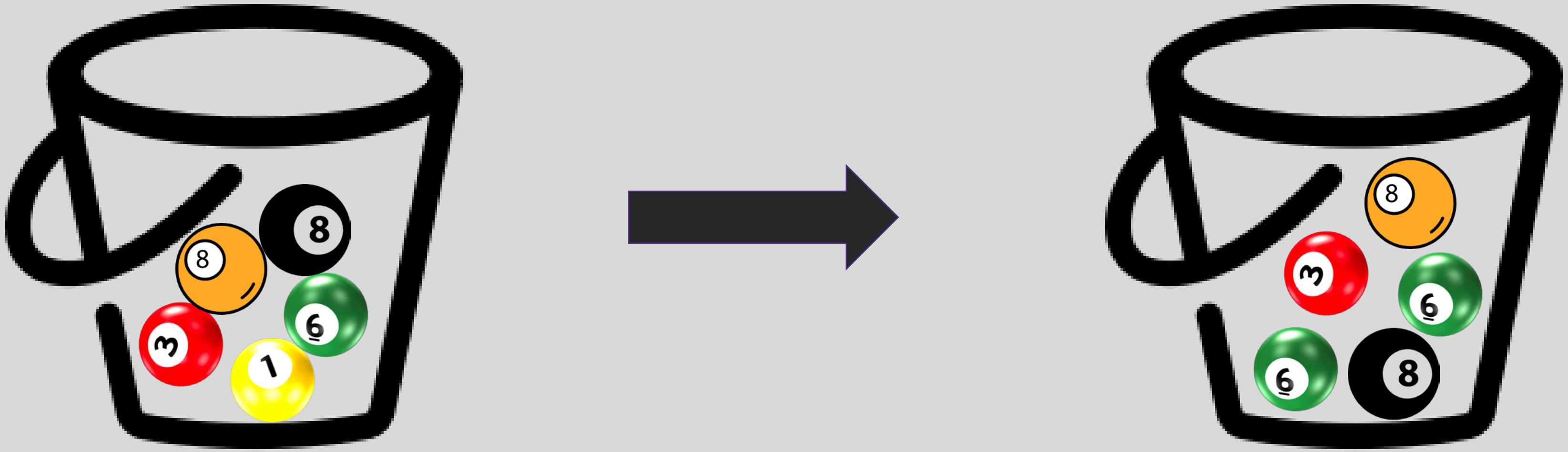
And again



# Bootstrap

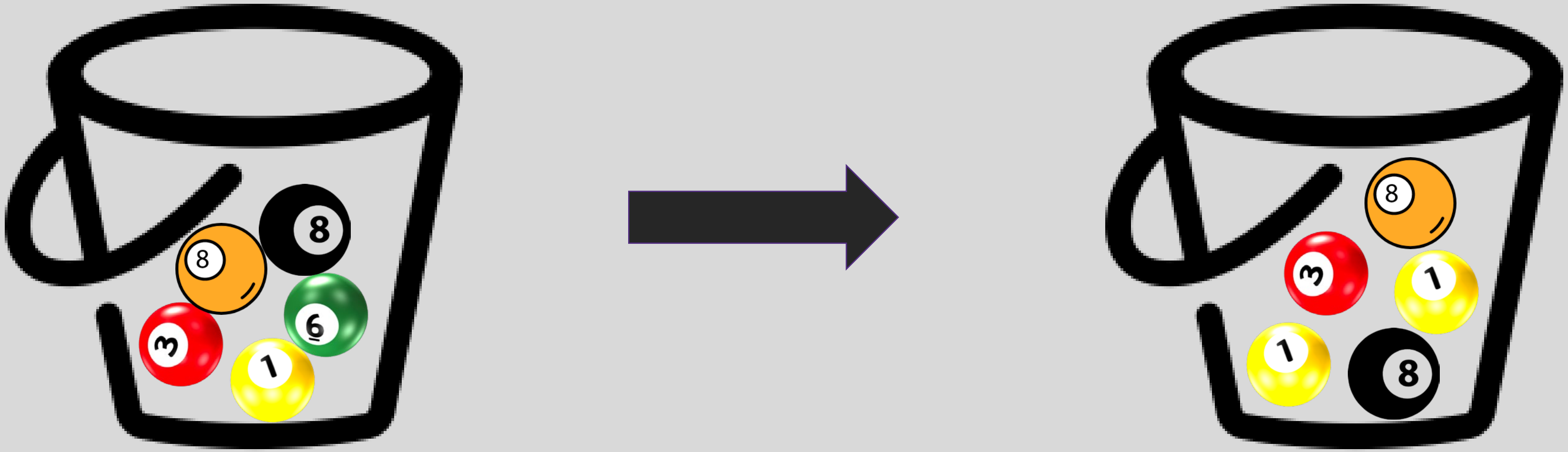
Until the "other" bucket has the same number of balls as the original one.




This new bucket represents a new parallel universe

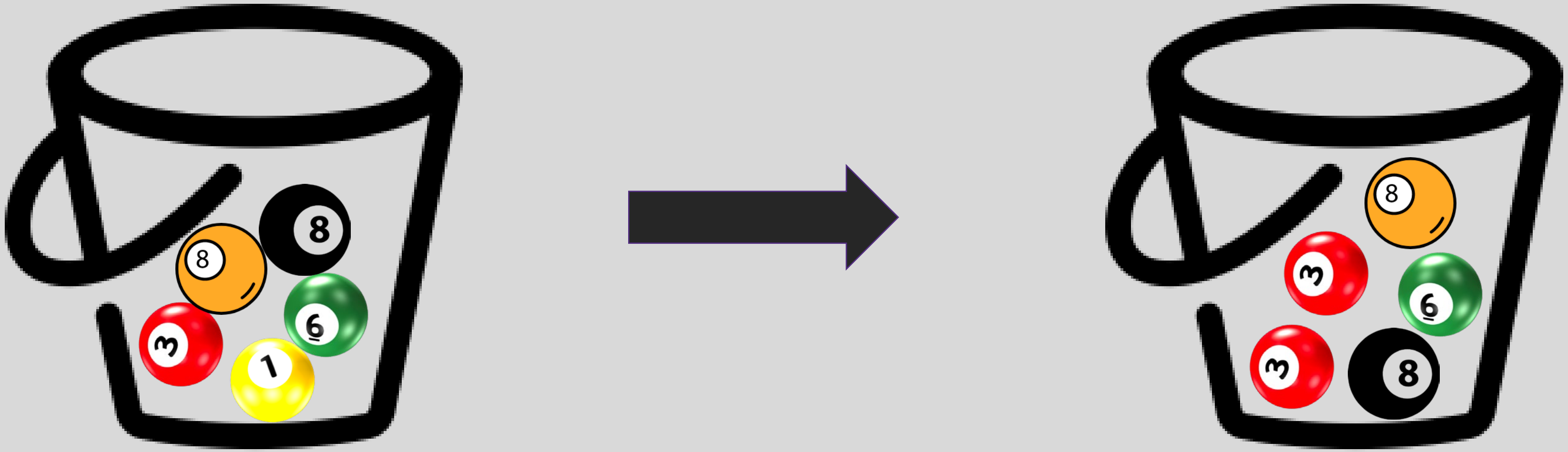
# Bootstrap

We repeat this same process and acquire another sample.



# Bootstrap

Until the "other" bucket has **the same number of balls** as the original one.




These new buckets represent the parallel universes

# Bootstrapping for Estimating Sampling Error

**Definition**

Bootstrapping is the practice of estimating properties of an estimator by measuring those properties by, for example, sampling from the observed data.

For example, we can compute $\hat{\beta}_0$ and $\hat{\beta}_1$ multiple times by randomly sampling from our data set. We then use the variance of our multiple estimates to approximate the true variance of $\hat{\beta}_0$ and $\hat{\beta}_1$.

# CIs for the predictors estimates: Standard Errors

We can empirically estimate the standard errors, $SE(\hat{\beta}_0), SE(\hat{\beta}_1)$ of $\beta_0$ and $\beta_1$ through bootstrapping.

If for each bootstrapped sample the estimated betas are: $\hat{\beta}_{0,i}$, $\hat{\beta}_{1,i}$, then

$$SE(\hat{\beta}_0) = \sqrt{\mathrm{var}(\widehat{\beta_0})}$$

$$SE(\hat{\beta}_1) = \sqrt{\mathrm{var}(\widehat{\beta_1})}$$

# CIs for the predictors estimates: Standard Errors

**Alternatively:**

If we know the variance $\sigma_\epsilon^2$ of the noise $\epsilon$, we can compute $SE(\hat{\beta}_0), SE(\hat{\beta}_1)$ analytically using the formulae below (no need to bootstrap):

$$SE\left(\widehat{\beta}_0\right) = \sigma_\epsilon \sqrt{\frac{1}{n} + \frac{\overline{x}^2}{\sum_i \left(x_i - \overline{x}\right)^2}}$$

$$SE\left(\widehat{\beta}_1\right) = \frac{\sigma_\epsilon}{\sqrt{\sum_i \left(x_i - \overline{x}\right)^2}}$$

# Standard Errors

**More data:** $n \uparrow$ and $\sum_i (x_i - \bar{x})^2 \uparrow \Longrightarrow SE \downarrow$

**Larger coverage:** $var(x)$ or $\sum_i (x_i - \bar{x})^2 \uparrow \Longrightarrow SE \downarrow$

**Better data:** $\sigma^2 \downarrow \Rightarrow SE \downarrow$

In practice, we do not know the theoretical value of $\sigma$ since we do not know the exact distribution of the noise $\epsilon$.

$$SE\left(\widehat{\beta}_0\right) = \sigma_\epsilon \sqrt{\frac{1}{n} + \frac{\overline{x}^2}{\sum_i \left(x_i - \overline{x}\right)^2}}$$

$$SE\left(\widehat{\beta}_1\right) = \frac{\sigma_\epsilon}{\sqrt{\sum_i \left(x_i - \overline{x}\right)^2}}$$

# Standard Errors

However, if we make the following assumptions,

- the errors $\epsilon_i = y_i - \hat{y}_i$ and $\epsilon_j = y_j - \hat{y}_j$ are uncorrelated, for $i \neq j$ ,
- each $\epsilon_i$ has a mean 0 and variance $\sigma_\epsilon^2$,

then, we can empirically estimate $\sigma^2$, from the data and our regression line:

Remember:

$$\boxed{\sigma_\epsilon \approx \sqrt{\frac{n \cdot \mathrm{MSE}}{n-2}} = \sqrt{\frac{\sum_i \left(y_i - \widehat{y}_i\right)^2}{n-2}}}$$

$$y_i = f(x_i) + \epsilon_i \Longrightarrow \epsilon_i = y_i - f(x_i)$$

# Standard Errors

**More data:** $n \uparrow$ and $\sum_i (x_i - \bar{x})^2 \uparrow \Longrightarrow SE \downarrow$

**Larger coverage:** $var(x)$ or $\sum_i (x_i - \bar{x})^2 \uparrow \Longrightarrow SE \downarrow$

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$$SE\left(\widehat{\beta}_0\right) = \sigma_\epsilon \sqrt{\frac{1}{n} + \frac{\overline{x}^2}{\sum_i \left(x_i - \overline{x}\right)^2}}$$

$$SE\left(\widehat{\beta}_1\right) = \frac{\sigma_\epsilon}{\sqrt{\sum_i \left(x_i - \overline{x}\right)^2}}$$

**Better model:** $(\hat{f} - y_i) \downarrow \Longrightarrow \sigma \downarrow \Longrightarrow SE \downarrow$

$$\sigma \approx \sqrt{\sum \frac{(\hat{f}(x) - y_i)^2}{n-2}}$$

**Question:** What happens to the $\widehat{\beta_0}$, $\widehat{\beta_1}$ under these scenarios?

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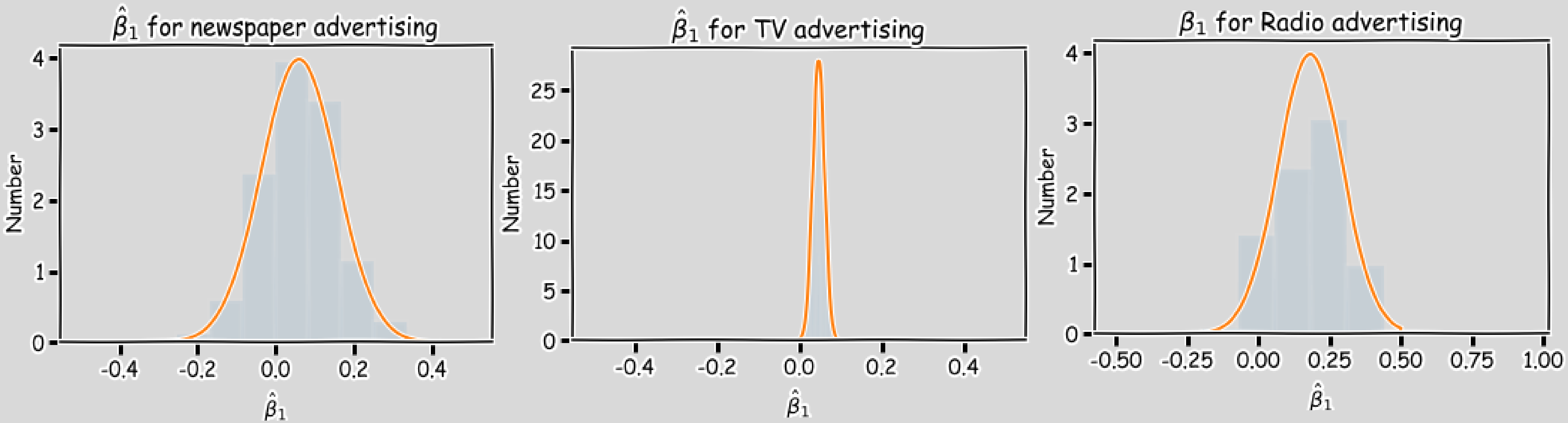
# Confidence intervals for the predictors estimates

And also we can answer the question, ’how significant are the predictors?’ Here we show the same analysis for all three predictors.

Question: Which ones are important?




Before we answer this question, we need to answer another question.

# Hypothesis Testing

Hypothesis testing is a formal process through which we evaluate the validity of a statistical hypothesis by considering evidence for or against the hypothesis gathered by ***random sampling of the data***.

1. State the hypotheses, typically a **null hypothesis**, $H_0$ and an **alternative hypothesis**, $H_1$, that is the negation of the former.
2. Choose a type of analysis, i.e. how to use sample data to evaluate the null hypothesis. Typically this involves choosing a single test statistic.
3. **Sample** data and compute the test statistic.
4. Use the value of the test statistic to either reject or not reject the null hypothesis.

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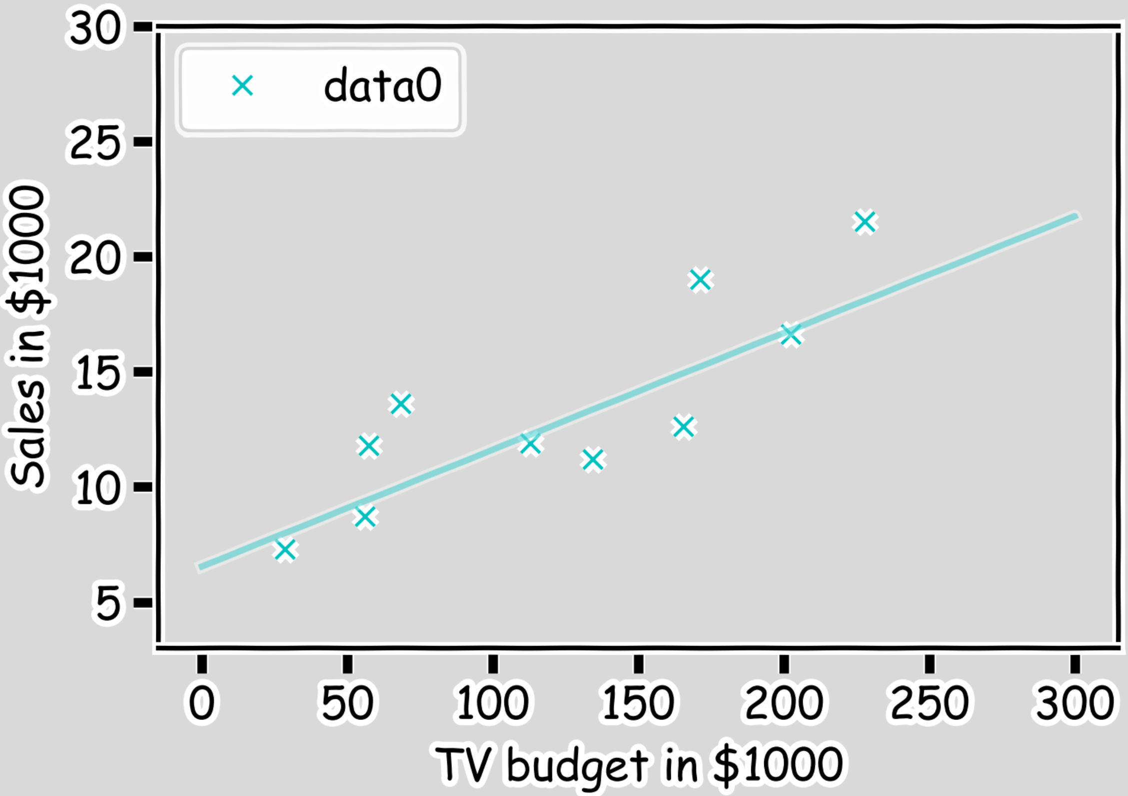
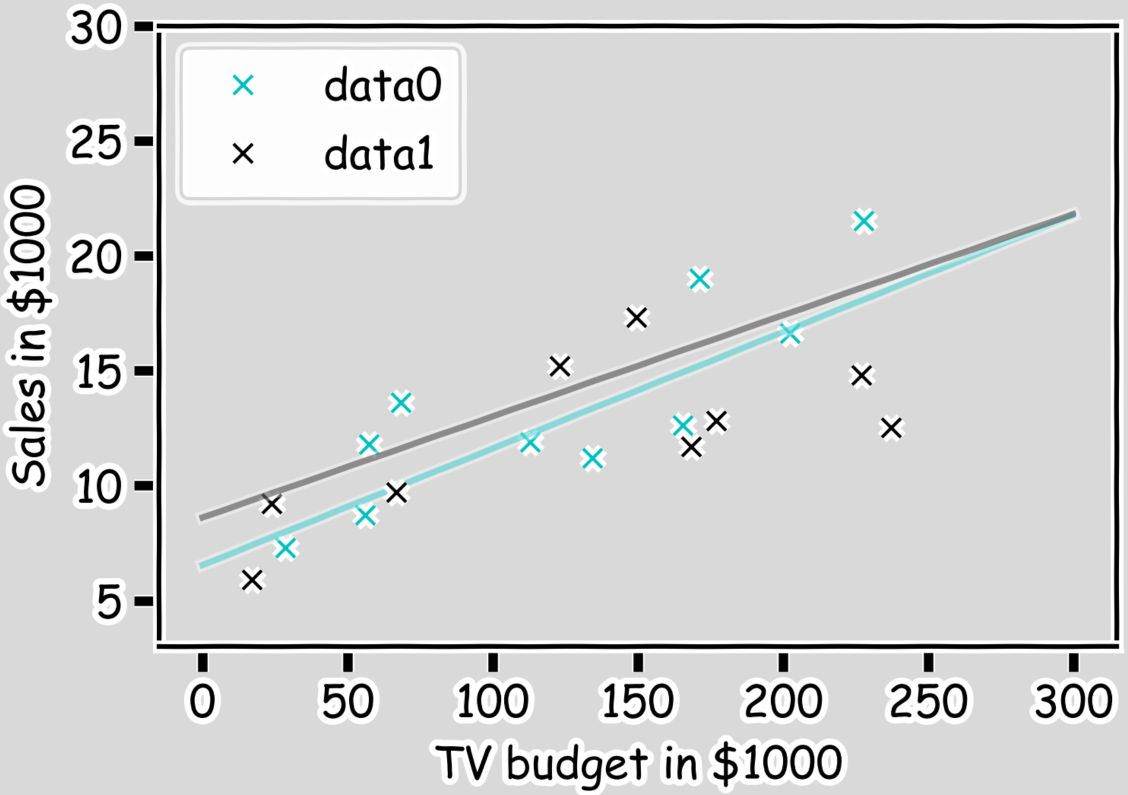
# How well we know the model $\hat{f}$?

Our confidence in $f$ is directly connected with the confidence in $\beta$s. So for each bootstrap sample, we have one $\beta$ which we can use to determine the model.



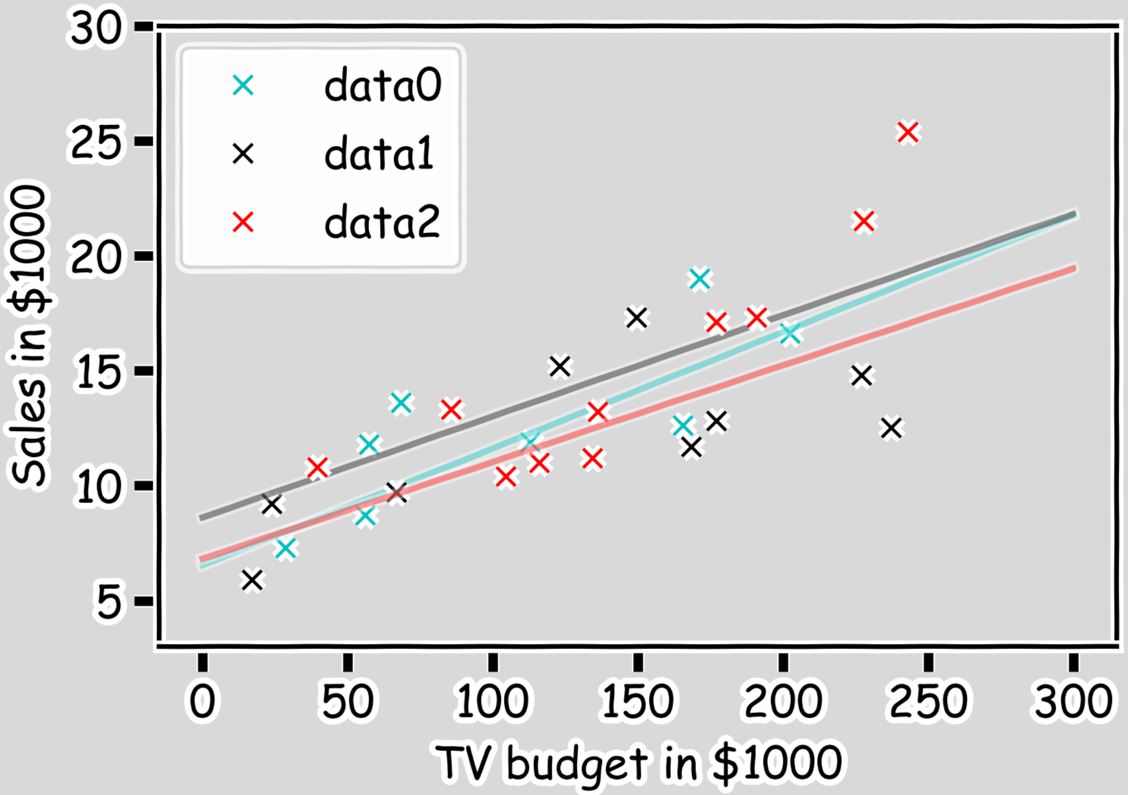
# How well we know the model $\hat{f}$?

Here we show two difference set of models given the fitted coefficients.



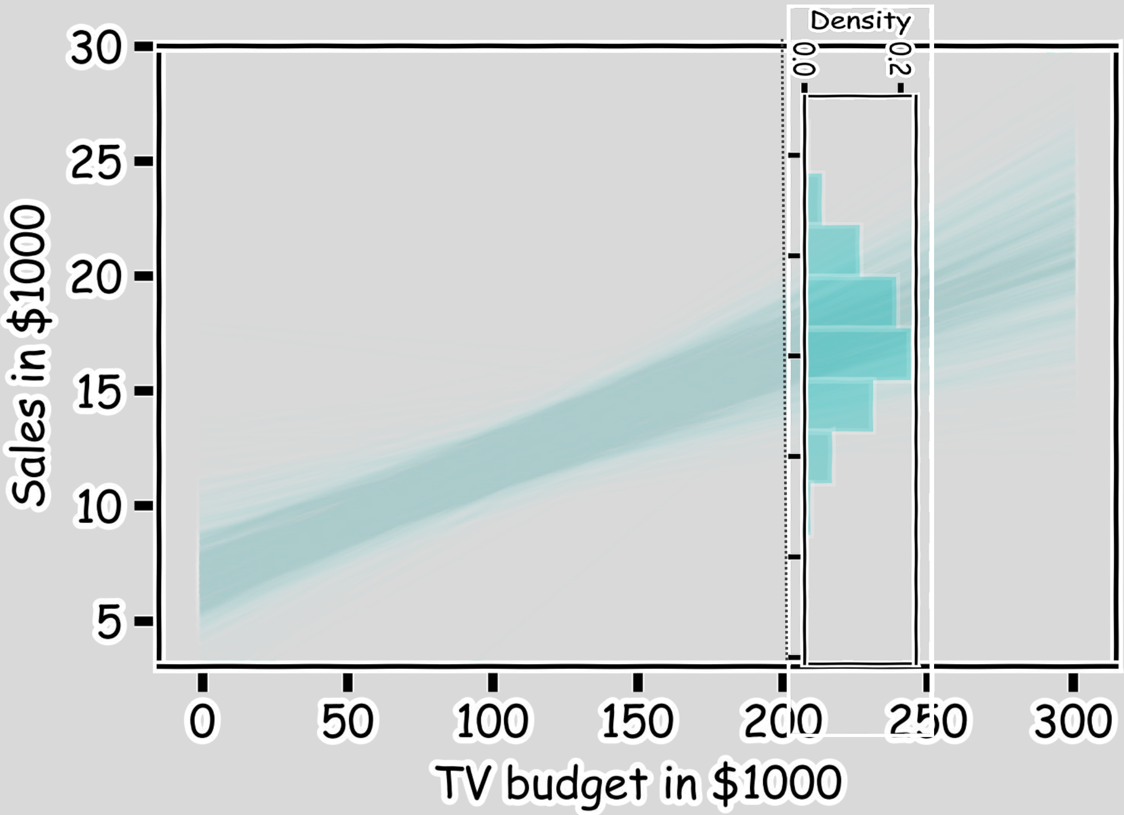
# How well we know the model $\hat{f}$?

There is one such regression line for every bootstrapped sample.



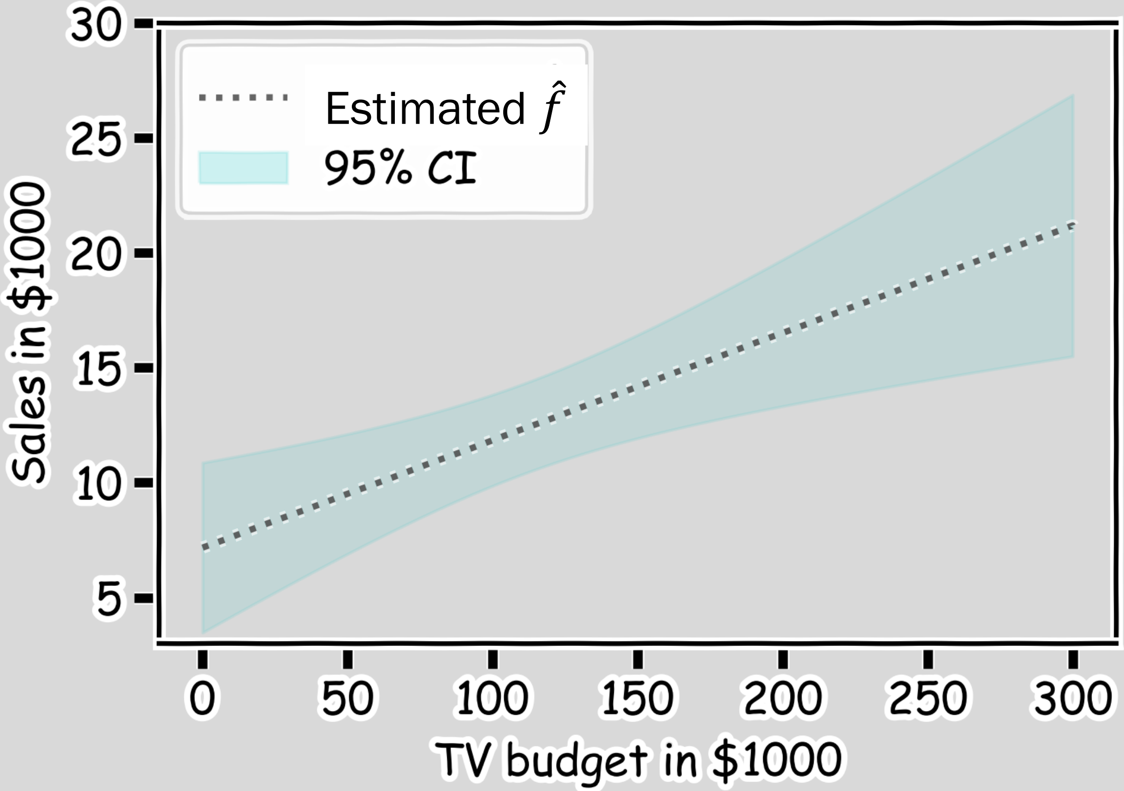
# How well we know the model $\hat{f}$?

Below we show all regression lines for a thousand of such bootstrapped samples.

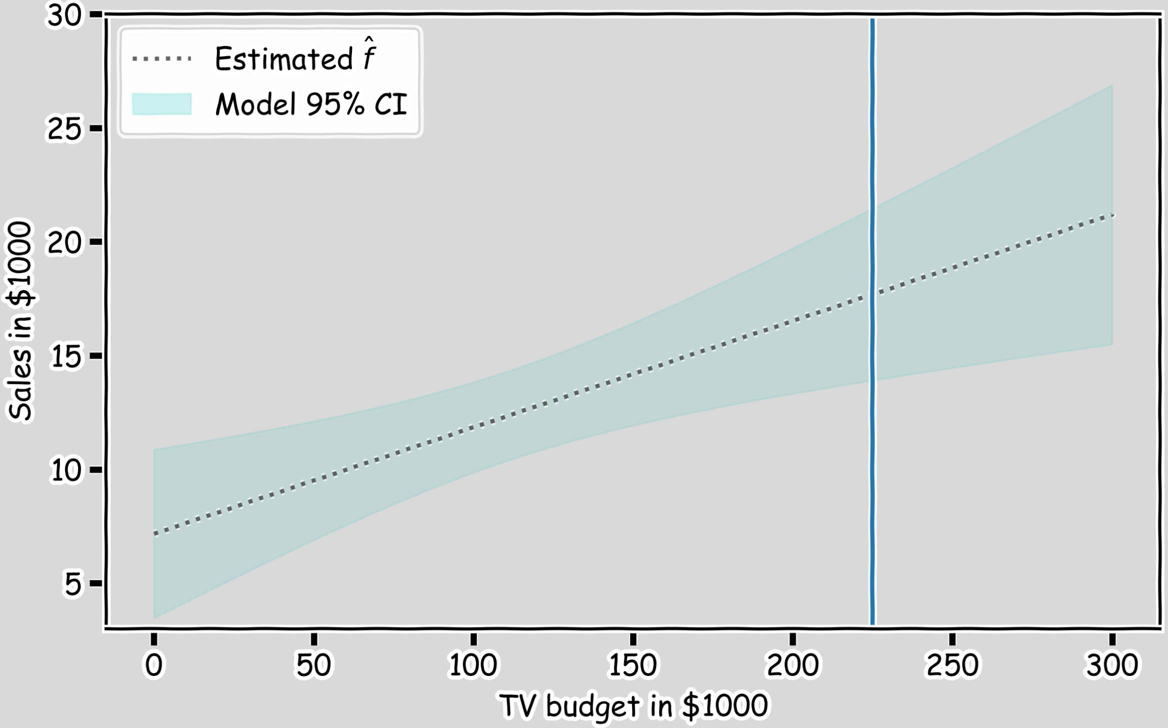


# How well we know the model $\hat{f}$?

For every $x$, we calculate the mean of the models, $\hat{f}$ (shown with dotted line) and the 95% CI of those models (shaded area).

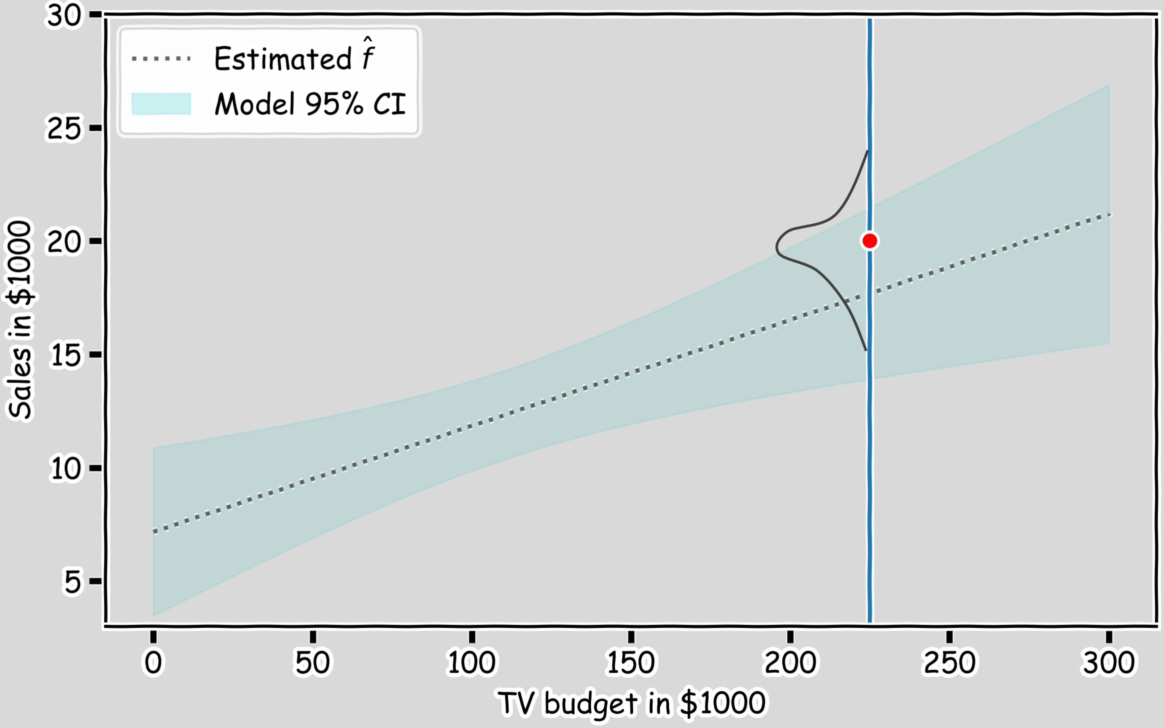


# Confidence in predicting $\hat{y}$



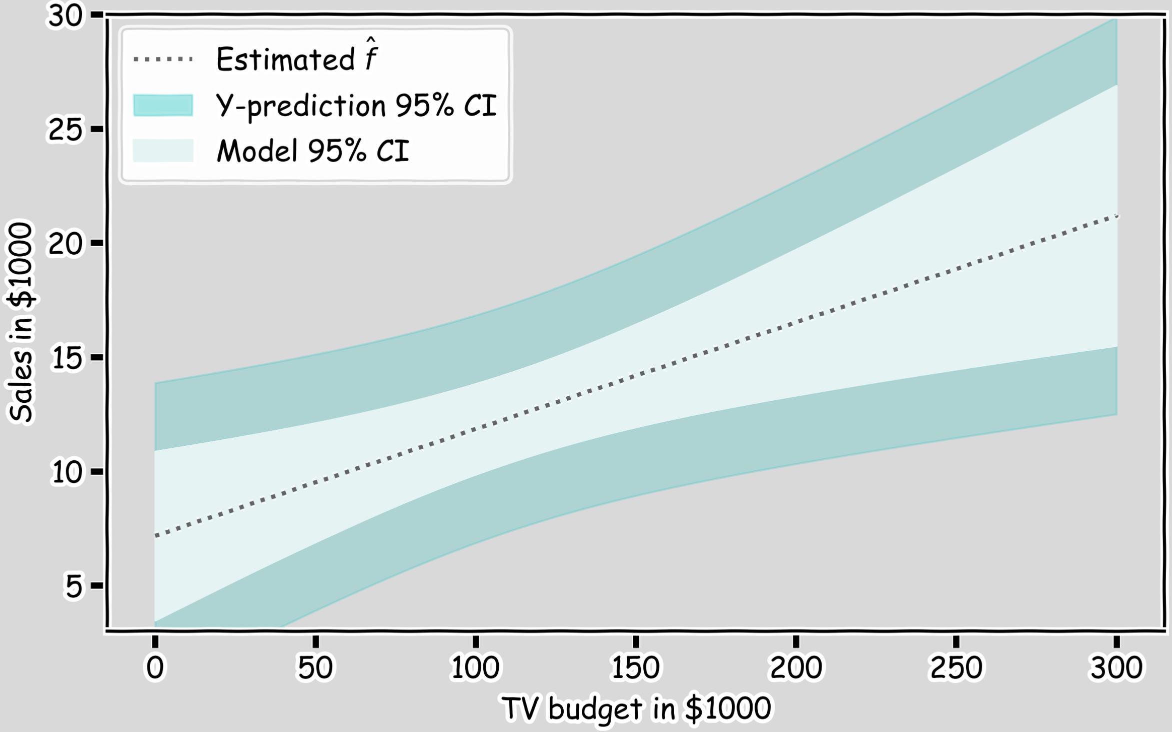
# Confidence in predicting $\hat{y}$

- for a given x, we have a distribution of models $f(x)$
- for each of these $f(x)$, the prediction for $y \sim N(f, \sigma_\epsilon)$



# Confidence in predicting $\hat{y}$

- for a given $x$, we have a distribution of models $f(x)$
- for each of these $f(x)$, the prediction for $y \sim N(f, \sigma_\epsilon)$
- The prediction confidence intervals are then



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# Questions?